Benchmarking AES on a Common Architecture

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Abstract:

This paper presents the design, implementation, and benchmarking of an AES-128 symmetric key encryption and decryption system on a Nexys A7 FPGA using a unified hardware datapath architecture. The design uses a single AES core to perform both encryption and decryption operations which are controlled by a mode-select signal. This ensures consistent evaluation of conditions across all performance metrics. The system was implemented in Verilog and synthesized using Vivado. Performance was evaluated in terms of latency, throughput, resource utilization, and power consumption. The design achieves a latency of 10 clock cycles per 128-bit block at a clock frequency of 100 MHz making a throughput of 1.28 Gbit/s. Resource usage remains low, with minimal LUT and flip-flop utilization and no reliance on DSP slices. Power analysis indicates efficient operation suitable for embedded and secure hardware applications. The results demonstrate that encryption and decryption have identical performance characteristics under the same architecture.

Key Terms - AES-128, FPGA, Hardware Datapath Architecture, Iterative Design, Symmetric Key Cryptography, Throughput, Latency, Resource Utilization, Power Analysis, Verilog, Nexys A7

I. Introduction

Advanced Encryption Standard (AES) is one of the most widely used symmetric key cryptographic algorithms for securing digital communication [2]. With the increasing demand for secure and high-speed data processing, hardware-based implementations of AES on Field Programmable Gate Arrays (FPGAs) [5] have gained significant importance due to their parallelism, flexibility, and performance efficiency [6].

This project focuses on implementing AES-128 encryption and decryption on a Nexys A7 FPGA using a unified hardware datapath architecture [7]. A single hardware design is used for both encryption and decryption, which ensures fair benchmarking and consistent performance evaluation. The system is analyzed based on latency, throughput, resource utilization, and power consumption.

II. Related Work

Previous research has explored various FPGA-based implementations of AES, including fully pipelined, partially pipelined, and iterative architectures. High-throughput designs achieve significant performance improvements at the cost of increased resource utilization. While the other compact forms of implementations focus on minimizing hardware usage.

Iterative architectures, such as the one used in this project, provide a balanced trade-off between performance and resource efficiency. Studies such as Hodjat and Verbauwhede [5] demonstrate high-speed AES implementations, while compact designs by Chodowiec and Gaj [10] focus on reduced hardware cost. This project adopts an iterative approach to maintain efficiency while enabling fair comparison between encryption and decryption using the same hardware.

III. Symmetric Key AES

AES is a symmetric block cipher that operates on 128-bit data blocks using keys of 128, 192, or 256 bits. In AES-128, the algorithm preforms 10 rounds of transformation with four steps each:

- SubBytes
- ShiftRows
- MixColumns
- AddRoundKey

Decryption performs the inverse operations in reverse order. The structure makes it well suited for hardware

implementation because of its simultaneously executable tasks.

IV. Design

The system is based on a unified hardware datapath architecture. A single AES core is designed to support both encryption and decryption operations using a control signal (`enc_dec`) to select the mode which in my implementation was sw[0].

Key design features:

- Round-based architecture (1 round per clock cycle)
- Shared datapath for encryption and decryption
- Finite State Machine (FSM) for control
- Internal key expansion logic
- Register-based state storage

This approach ensures:

- consistent conditions for benchmarking
- reduced hardware duplication
- efficient resource utilization

V. Implementation on FPGA

The proposed AES-128 encryption and decryption system was implemented on a Xilinx Nexys A7 FPGA platform using Verilog HDL. The design was developed and synthesized using Xilinx Vivado design suite. A unified datapath architecture was employed, allowing both encryption and decryption operations to be executed using a single AES core. The selection between encryption and decryption modes is controlled through a mode-select signal, ensuring that both operations share identical hardware resources and execution conditions.

The design follows a repetitive architecture, where one round of the AES algorithm is executed per clock cycle. This approach reduces hardware resource utilization compared to fully pipelined designs while maintaining an acceptable performance. The system operates at a clock frequency of 100 MHz, with synchronous design practices applied throughout the system. Key expansion, round transformation, and control logic were integrated into a single module to streamline the datapath. The implementation avoids the use of DSP slices, relying instead on lookup tables (LUTs) and flip-flops, making it suitable for resource-constrained FPGA environments.

VI. Results

The system was verified using standard AES test vectors. The encryption output matched the expected ciphertext, and decryption correctly recovered the original plaintext.

```
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
```

This is the console message in Vivado which provides information for cycles/block for both encryption and decryption.

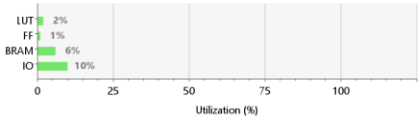
```
ENC done in 10 cycles
ENC result = 69c4e0d86a7b0430d8cdb78070b4c55a
DEC done in 10 cycles
DEC result = 00112233445566778899aabbccddeeff
```

Performance metrics:

Case	Clock (MHZ)	Cycles/Block	Throughput (Gbit/s)	LUT	FF	BRAM	DSP	Power (W)
AES-128 Encrypt	100	10	1.28	953	168	8	0	0.135
AES-128 Decrypt	100	10	1.28	953	168	8	0	0.135

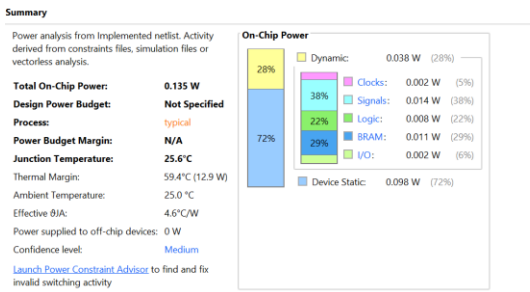
a. Resource Utilization

Resource	Utilization	Available	Utilization %
LUT	953	63400	1.50
FF	168	126800	0.13
BRAM	8	135	5.93
IO	20	210	9.52



The absence of DSP usage confirms that AES operations were implemented using logic-based transformations rather than arithmetic multipliers.

b. Power Usage



This demonstrates that the design was energy efficient and could be used for embedded systems.

c. Comparisons

Since both encryption and decryption use the same hardware architecture:

- Latency is identical (10 cycles)
- Throughput is identical (1.28 Gbit/s)
- Resource utilization is identical
- Power consumption is nearly identical

This validates the fairness of the benchmarking methodology. Since both encryption and decryption go through the same steps and number of transformations, it should remain around the same for all fields tested.

VII. Applications

Our implementation of AES-128 is primarily designed for use with devices such as FPGAs, embedded systems, and CPUs to provide efficient and reliable encryption. AES-128 is heavily used in transport layer security (TLS), which powers HTTPS and affects browsing, logging into accounts, and making online payments. Since it is the smallest key size for the AES algorithm, it excels in speed and provides lower latency than the larger key sizes. It can be used as miniature cryptography hardware for routers, phones, servers, and more.

VIII. Conclusion

This project successfully demonstrates the implementation and benchmarking of AES-128 on FPGA using the same architecture. The design achieves high throughput with low resource utilization and power consumption. Through using the same architecture for both encryption and decryption it ensures a fair evaluation of performance metrics on AES-128 for symmetric key algorithms. The results demonstrate the effectiveness of FPGA-based AES implementations for hardware applications.

References

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